RAJALAKSHMI ENGINEERING COLLEGE
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CS23431 - OPERATING SYSTEM

EXP 4: Round Robin Algorithm Using C

QUESTION :

Write a C program to implement the Round Robin (RR) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT), Burst Times (BT), and a Time Quantum (TQ). The CPU should rotate through the processes in the ready queue, granting each a maximum of one TQ of execution time. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and the overall averages.

CODE:

```
#include <stdio.h>
int main() {
    int n, tq, i;
    printf("Enter number of processes: \n");
    scanf("%d", &n);
    printf("Enter Time Quantum: \n");
    scanf("%d", &tq);
    int AT[n], BT[n], RT[n], CT[n], TAT[n], WT[n];
    for(i = 0; i < n; i++) {
        printf("Enter AT and BT for P%d: \n", i + 1);
        scanf("%d %d", &AT[i], &BT[i]);
        RT[i] = BT[i];
    }
    int completed = 0, current_time = 0;
    float totalTAT = 0, totalWT = 0;
    while(completed < n) {
        int executed = 0;
        for(i = 0; i < n; i++) {
            if(AT[i] <= current_time && RT[i] > 0) {
                executed = 1;
                if(RT[i] > tq) {
                    current_time += tq;
                    RT[i] -= tq;
                } else {
                    current_time += RT[i];
                    RT[i] = 0;
                    CT[i] = current_time;
                    completed++;
```

```
                    }
                }
            }
            if(executed == 0)
                current_time++;
        }
        for(i = 0; i < n; i++) {
            TAT[i] = CT[i] - AT[i];
    }
```

Output:

Custom Test — Success

Input :

```
3
2
0 5
1 3
2 1
```

Expected Output :

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
```

Output :

```
Enter number of processes:
Enter Time Quantum:
Enter AT and BT for P1:
Enter AT and BT for P2:
Enter AT and BT for P3:

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
```